

Spur Gears Component Generator (Version: 2026.2 (Build 302298000, 298))

12/17/2025

Project Info (iProperties)

Guide

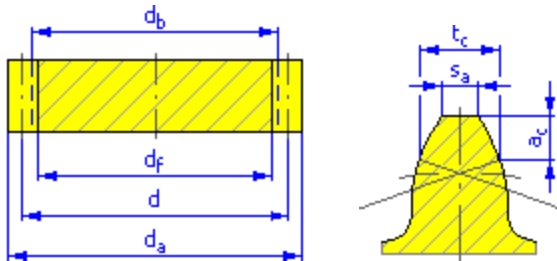
Common Parameters

Gear Ratio	i	4.1200 ul
Desired Gear Ratio	i _{in}	4.1200 ul
Module	m	2.000 mm
Helix Angle	β	10.0633 deg
Pressure Angle	α	20.0000 deg
Center Distance	a _w	130.000 mm
Product Center Distance	a	130.000 mm
Total Unit Correction	Σx	0.0000 ul
Circular Pitch	p	6.283 mm
Base Circular Pitch	p _{tb}	5.986 mm
Operating Pressure Angle	α _w	20.0000 deg
Tangential Pressure Angle	α _t	20.2872 deg
Tangential Operating Pressure Angle	α _{tw}	20.2872 deg
Base Helix Angle	β _b	9.4506 deg
Tangential Module	m _t	2.031 mm
Tangential Circular Pitch	p _t	6.381 mm
Contact Ratio	ε	3.1689 ul
Transverse Contact Ratio	ε _α	1.6950 ul
Overlap Ratio	ε _β	1.4739 ul
Limit Deviation of Axis Parallelity	f _x	0.0280 mm
Limit Deviation of Axis Parallelity	f _y	0.0140 mm

Gears

		Gear 1	Gear 2
Type of model		Component	Component
Number of Teeth	z	25 ul	103 ul
Unit Correction	x	0.0000 ul	0.0000 ul
Pitch Diameter	d	50.781 mm	209.219 mm
Outside Diameter	d _a	54.781 mm	213.219 mm
Root Diameter	d _f	45.781 mm	204.219 mm
Base Circle Diameter	d _b	47.631 mm	196.240 mm
Work Pitch Diameter	d _w	50.781 mm	209.219 mm
Facewidth	b	56.000 mm	53.000 mm
Facewidth Ratio	b _r	1.0437 ul	0.2533 ul
Addendum	a*	1.0000 ul	1.0000 ul

Clearance	c^*	0.2500 ul	0.2500 ul
Root Fillet	r_f^*	0.3500 ul	0.3500 ul
Tooth Thickness	s	3.142 mm	3.142 mm
Tangential Tooth Thickness	s_t	3.191 mm	3.191 mm
Chordal Thickness	t_c	2.774 mm	2.774 mm
Chordal Addendum	a_c	1.495 mm	1.495 mm
Chordal Dimension	W	21.397 mm	76.819 mm
Chordal Dimension Teeth	z_w	4.000 ul	13.000 ul
Dimension Over (Between) Wires	M	55.639 mm	214.255 mm
Wire Diameter	d_M	3.500 mm	3.500 mm
Limit Deviation of Helix Angle	F_β	0.0280 mm	0.0290 mm
Limit Circumferential Run-out	F_r	0.0420 mm	0.0550 mm
Limit Deviation of Axial Pitch	f_{pt}	0.0150 mm	0.0170 mm
Limit Deviation of Basic Pitch	f_{pb}	0.0140 mm	0.0160 mm
Virtual Number of Teeth	z_v	26.094 ul	107.508 ul
Virtual Pitch Diameter	d_n	52.188 mm	215.016 mm
Virtual Outside Diameter	d_{an}	56.188 mm	219.016 mm
Virtual Base Circle Diameter	d_{bn}	49.041 mm	202.049 mm
Unit Correction without Tapering	x_z	0.2902 ul	-2.1123 ul
Unit Correction without Undercut	x_p	-0.5065 ul	-5.2683 ul
Unit Correction Allowed Undercut	x_d	-0.6765 ul	-5.4383 ul
Addendum Truncation	k	0.0000 ul	0.0000 ul
Unit Outside Tooth Thickness	s_a	0.7214 ul	0.8087 ul
Tip Pressure Angle	α_a	29.4158 deg	22.7703 deg



Loads

		Gear 1	Gear 2
Power	P	5.200 kW	5.050 kW
Speed	n	2580.00 rpm	626.21 rpm
Torque	T	19.246 N m	77.002 N m
Efficiency	η	0.971 ul	
Radial Force	F_r	280.199 N	
Tangential Force	F_t	757.996 N	
Axial Force	F_a	134.518 N	
Normal Force	F_n	819.247 N	
Circumferential Speed	v	6.860 mps	
Resonance Speed	n_{E1}	19094.860 rpm	

Material

		Gear 1	Gear 2
		42CrV6	42CrV6
Ultimate Tensile Strength	S_u	980 MPa	980 MPa
Yield Strength	S_y	850 MPa	850 MPa
Modulus of Elasticity	E	206000 MPa	206000 MPa
Poisson's Ratio	μ	0.300 ul	0.300 ul
Bending Fatigue Limit	σ_{Flim}	700.0 MPa	530.0 MPa
Contact Fatigue Limit	σ_{Hlim}	1160.0 MPa	720.0 MPa
Hardness in Tooth Core	JHV	210 ul	210 ul
Hardness in Tooth Side	VHV	600 ul	600 ul
Base Number of Load Cycles in Bending	N_{Flim}	3000000 ul	3000000 ul
Base Number of Load Cycles in Contact	N_{Hlim}	100000000 ul	50000000 ul
Wöhler Curve Exponent for Bending	q_F	6.0 ul	6.0 ul
Wöhler Curve Exponent for Contact	q_H	10.0 ul	10.0 ul
Type of Treatment	type	2 ul	0 ul

Strength Calculation

Factors of Additional Load

Application Factor	K_A	1.000 ul	
Dynamic Factor	K_{Hv}	1.557 ul	1.557 ul
Face Load Factor	$K_{H\beta}$	1.400 ul	1.400 ul
Transverse Load Factor	K_{Ha}	1.400 ul	1.400 ul
One-time Overloading Factor	K_{AS}	1.000 ul	

Factors for Contact

Elasticity Factor	Z_E	189.812 ul	
Zone Factor	Z_H	2.471 ul	
Contact Ratio Factor	Z_ϵ	0.762 ul	
Single Pair Tooth Contact Factor	Z_B	1.000 ul	1.000 ul
Life Factor	Z_N	0.994 ul	0.984 ul
Lubricant Factor	Z_L	0.965 ul	
Roughness Factor	Z_R	0.783 ul	
Speed Factor	Z_v	0.984 ul	
Helix Angle Factor	Z_β	0.994 ul	
Size Factor	Z_X	1.000 ul	1.000 ul
Work Hardening Factor	Z_W	1.000 ul	

Factors for Bending

Form Factor	Y_{Fa}	2.564 ul	2.175 ul
Stress Correction Factor	Y_{Sa}	1.632 ul	1.848 ul
Teeth with Grinding Notches Factor	Y_{Sag}	1.000 ul	1.000 ul

Helix Angle Factor	Y_{β}	0.928 ul	
Contact Ratio Factor	Y_{ϵ}	0.676 ul	
Alternating Load Factor	Y_A	1.000 ul	1.000 ul
Production Technology Factor	Y_T	1.000 ul	1.000 ul
Life Factor	Y_N	1.000 ul	1.000 ul
Notch Sensitivity Factor	Y_{δ}	1.065 ul	1.073 ul
Size Factor	Y_X	1.000 ul	1.000 ul
Tooth Root Surface Factor	Y_R	1.000 ul	

Results

Factor of Safety from Pitting	S_H	2.336 ul	1.434 ul
Factor of Safety from Tooth Breakage	S_F	13.751 ul	10.335 ul
Static Safety in Contact	S_{Hst}	6.444 ul	6.444 ul
Static Safety in Bending	S_{Fst}	32.278 ul	24.080 ul
Check Calculation		Positive	

Summary of Messages

11:28:48 AM Design: Gear 1: The Unit Correction (x) is less than the Unit Correction without Tapering (x₂)

11:28:48 AM Calculation: Calculation indicates design compliance!

Roller Chains Generator (Version: 2026.2 (Build 302298000, 298))

12/17/2025

Project Info (iProperties)

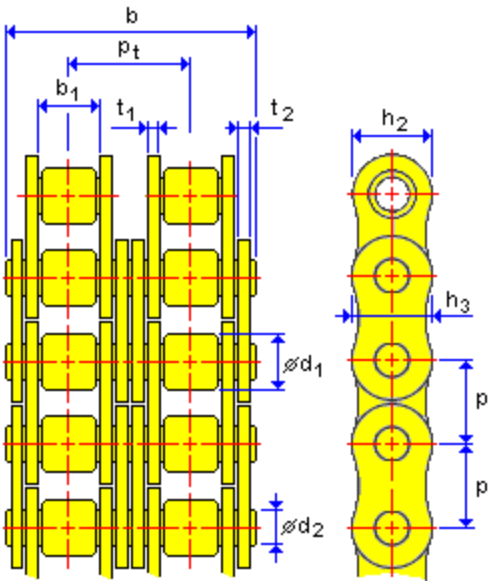
Summary

Project

Status

Chain properties

Chain : ISO 606:2004 - Short-pitch transmission precision roller chains (EU)		
Chain size designation	12B-1-124	
Pitch	p	19.050 mm
Number of Chain Links	X	124.000 ul
Number of Chain Strands	k	1.000 ul
Minimum width between inner plates	b ₁	11.680 mm
Maximum Roller Diameter	d ₁	12.070 mm
Maximum pin body diameter	d ₂	5.720 mm
Maximum inner plate depth	h ₂	16.130 mm
Maximum outer or intermediate plate depth	h ₃	16.130 mm
Maximum width over bearing pins	b	22.700 mm
Maximum inner plate width	t ₁	1.800 mm
Maximum outer or intermediate plate width	t ₂	1.700 mm
Chain bearing area	A	89.000 mm^2
Tensile Strength	F _u	28900.000 N
Specific Chain Mass	m	1.200 kg/m
Chain construction factor	φ	1.000 ul



Sprocket 1 properties: Toothed sprocket

Sprocket 2 properties: Toothed sprocket

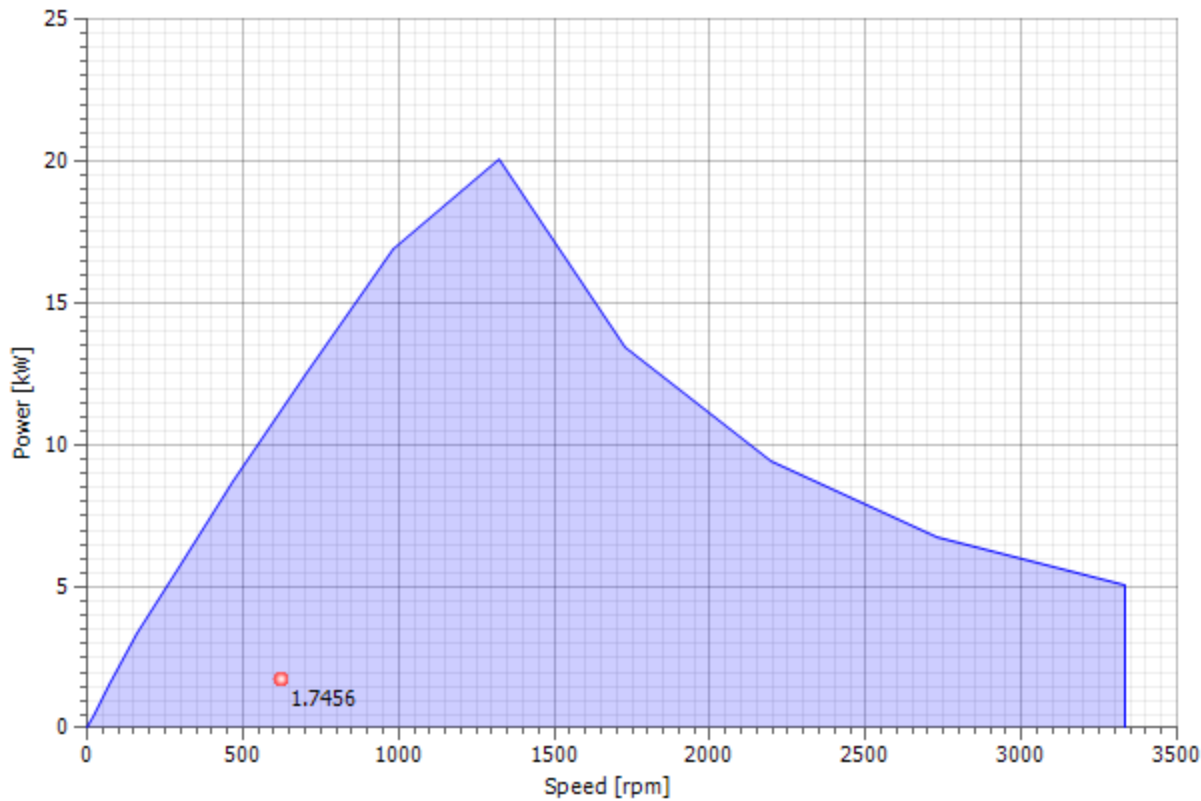
Working conditions

Power	P	5.050 kW
Torque	T	77.039 N m
Speed	n	626.000 rpm
Efficiency	η	0.965 ul
Required service life	L _h	867.240 hr
Maximum chain elongation	ΔL _{max}	0.030 ul
Application	Smooth running	
Environment	Clean	
Lubrication	Recommended (see notes below)	

Power correction factors

Shock factor	Y	1.000 ul
Service factor	f ₁	1.000 ul
Sprocket size factor	f ₂	1.000 ul
Strands factor	f ₃	1.000 ul
Lubrication factor	f ₄	1.000 ul
Center distance factor	f ₅	0.987 ul
Ratio factor	f ₆	1.095 ul
Service life factor	f ₇	0.320 ul

Chain power rating



Results

Chain Speed	v	4.189 mps
Effective pull	F _p	1205.462 N
Centrifugal force	F _C	21.062 N
Maximum tension in chain span	F _{Tmax}	1226.524 N
Static safety factor	S _S > S _{Smin}	23.563 ul > 15.280 ul
Dynamic safety factor	S _D > S _{Dmin}	23.563 ul > 14.750 ul
Bearing pressure	p _B < p ₀ * λ	13.781 MPa
Permissible bearing pressure	p ₀	19.775 MPa
Specific friction factor	λ	2.893 ul
Design power	P _D < P _R	1.746 kW
Chain power rating	P _R	11.273 kW
Chain service life for specified elongation	t _h > L _h	160044 hr
Chain link plates service life	t _{hL} > L _h	2777778 hr

Roller and bushing service life	$t_{hr} > L_h$	101423 hr
---------------------------------	----------------	-----------

Summary of Messages

11:28:16 AM Calculation: Recommended lubrication : Oil bath or disc lubrication
11:28:16 AM Calculation: Maximum chain power rating is limited by link plate fatigue.
11:28:16 AM Calculation: Calculation indicates design compliance!

Shaft Component Generator (Version: 2026.2 (Build 302298000, 298))

12/17/2025

Project Info (iProperties)

Calculation

Material

Material		C45
Modulus of Elasticity	E	210000 MPa
Modulus of Rigidity	G	80000 MPa
Density	ρ	7800 kg/m^3

Calculation Properties

Include		
Yes	Density	ρ 7800 kg/m^3
Yes	Shear Displacement Ratio	β 1.188 ul
	Number of Divisions	1000 ul
	Mode of reduced stress	HMH

Loads

Index	Location	Radial Force				Bending Moment				Continuous Load				Axial Force	Torque	Deflection				Deflection Angle
		Y	X	Size	Direction	Y	X	Size	Direction	Y	X	Size	Direction			Y	X	Size	Direction	
1	24.75 mm	548.380 N		548.380 N												-8.069 microm	1.118 microm	8.146 microm	172.11 deg	0.01 deg
2	24.75 mm														19.246 N m	-8.069 microm	1.118 microm	8.146 microm	172.11 deg	0.01 deg
3	24.75 mm					3.415 N m		3.415 N m								-8.069 microm	1.118 microm	8.146 microm	172.11 deg	0.01 deg
4	115 mm														-19.246 N m	0.590 microm	-0.842 microm	1.028 microm	305.01 deg	0.00 deg
5	115 mm	286.560 N		286.560 N												0.590 microm	-0.842 microm	1.028 microm	305.01 deg	0.00 deg
6	115 mm	0.000 N	758.010 N	758.010 N	90.00 deg											0.590 microm	-0.842 microm	1.028 microm	305.01 deg	0.00 deg
7	115 mm													134.520 N		0.590 microm	-0.842 microm	1.028 microm	305.01 deg	0.00 deg

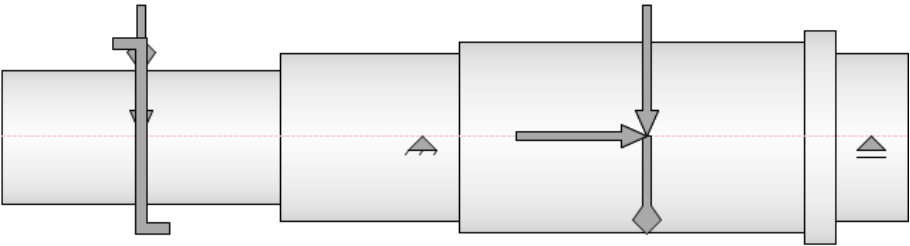
Supports

Index	Type	Location	Reaction Force					Yielding	Type	Deflection				Deflection Angle
			Y	X	Size	Direction	Axial Force			Y	X	Size	Direction	
1	Fixed	75 mm	1085.861 N	379.005 N	1150.104 N	19.24 deg	134.520 N		User	-0.000 microm	-0.000 microm	0.000 microm	180.00 deg	0.00 deg
2	Free	155 mm	-242.026 N	379.005 N	449.691 N	122.56 deg			User	-0.000 microm	-0.000 microm	0.000 microm	195.93 deg	0.00 deg

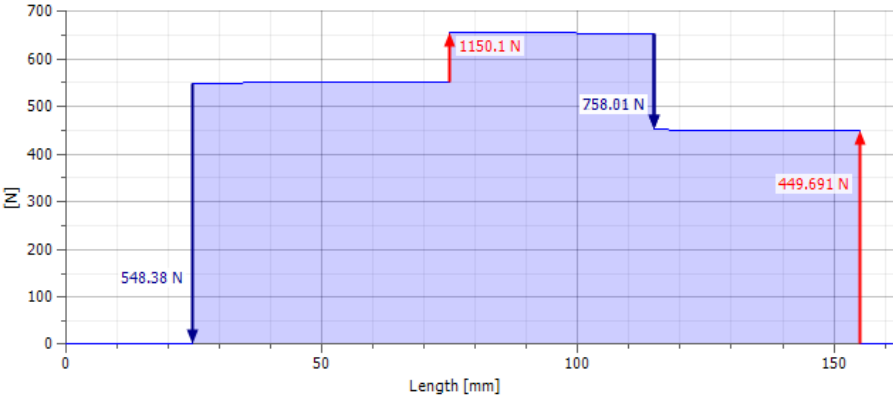
Results

Length	L	161.500 mm
Mass	Mass	0.907 kg
Maximal Bending Stress	σ _B	12.547 MPa
Maximal Shear Stress	τ _S	1.216 MPa
Maximal Torsional Stress	τ	7.091 MPa
Maximal Tension Stress	σ _T	0.190 MPa
Maximal Reduced Stress	σ _{red}	17.683 MPa
Maximal Deflection	f _{max}	13.480 microm
Angle of Twist	φ	0.02 deg

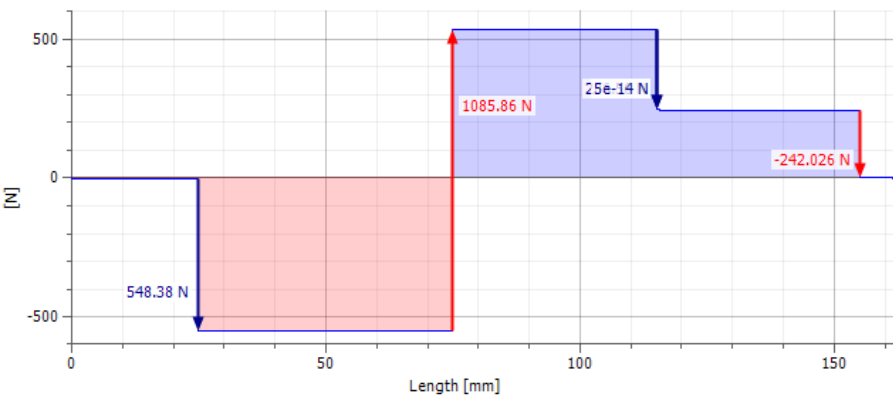
Preview



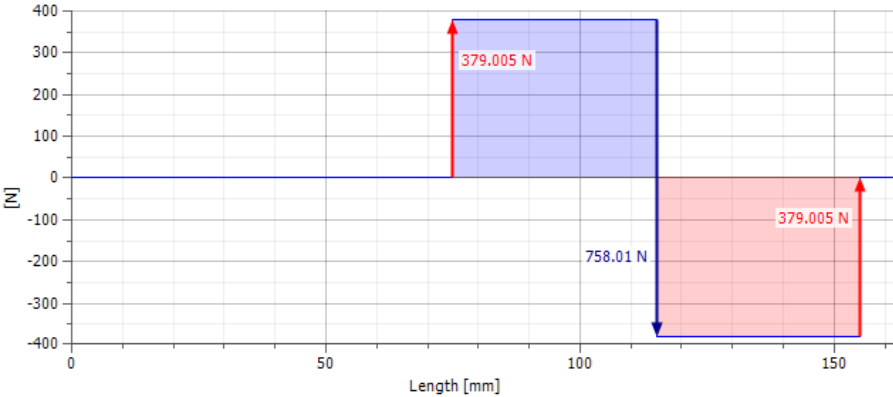
Shear Force



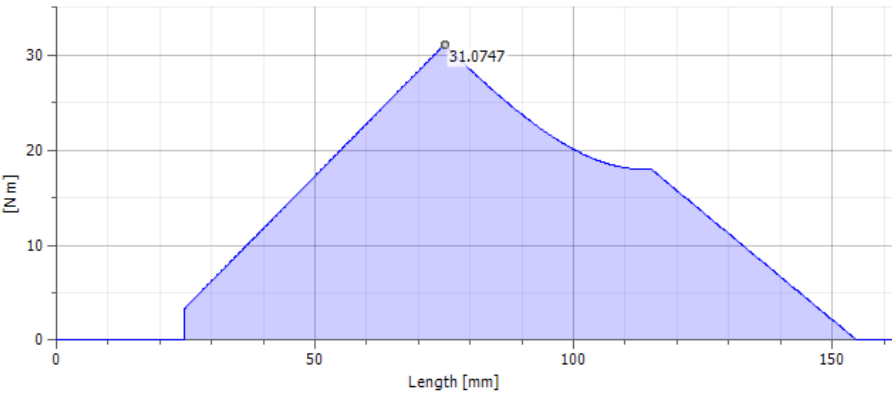
Shear Force, YZ Plane



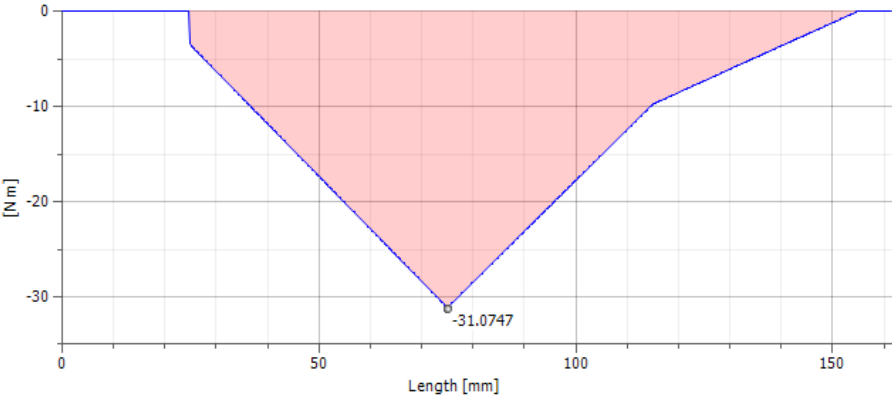
Shear Force, XZ Plane



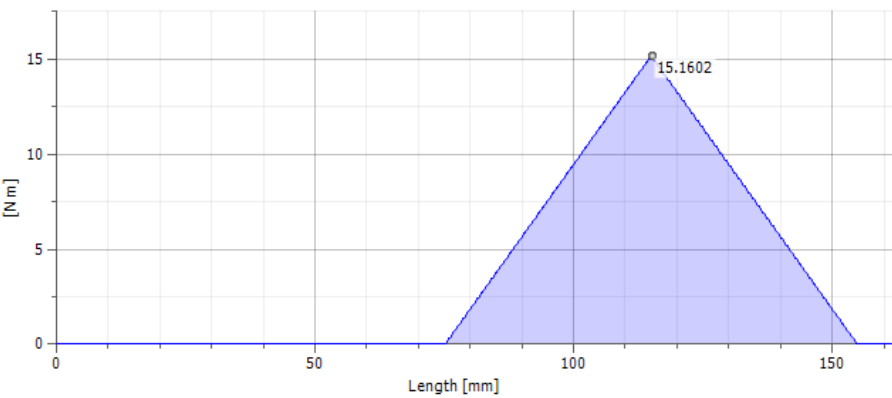
Bending Moment



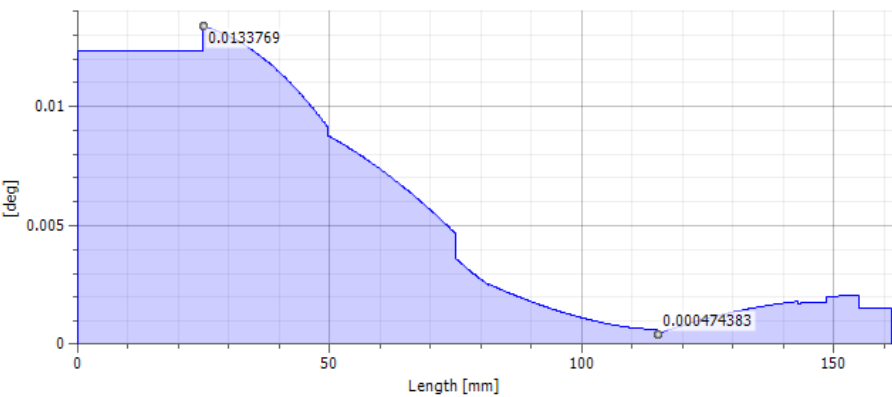
Bending Moment, YZ Plane



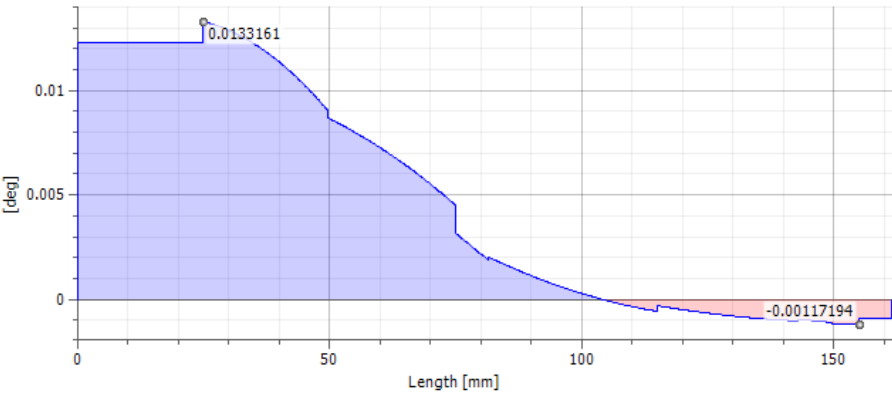
Bending Moment, XZ Plane



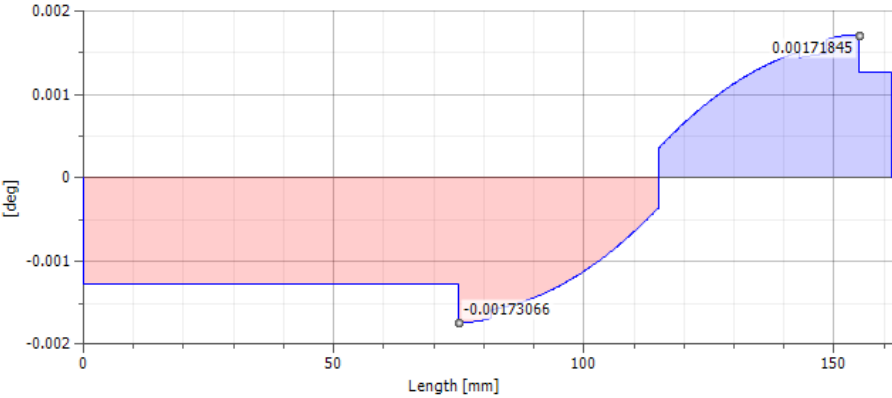
Deflection Angle



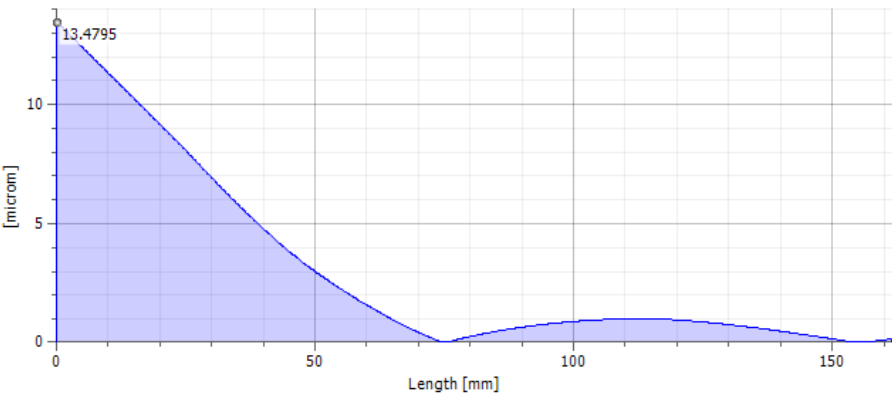
Deflection Angle, YZ Plane



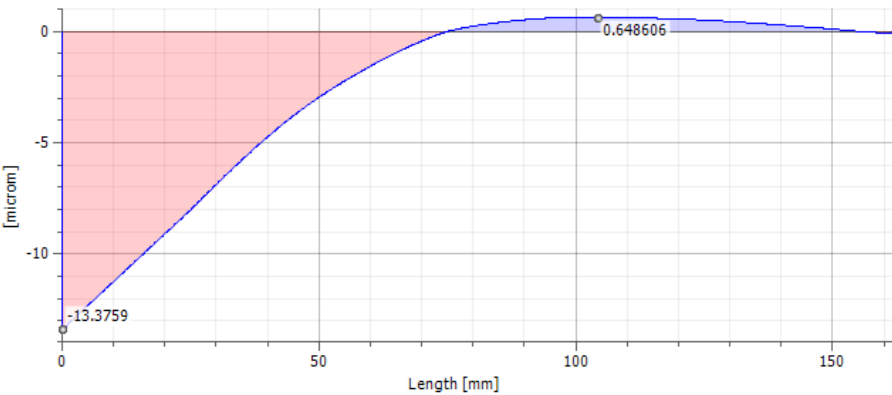
Deflection Angle, XZ Plane



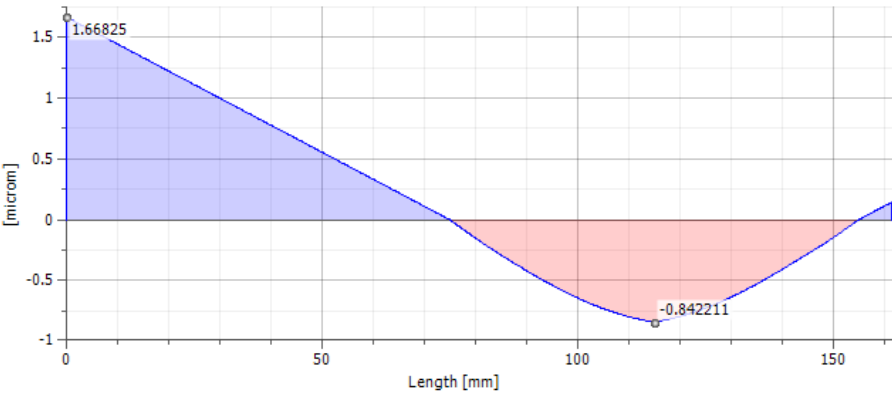
Deflection



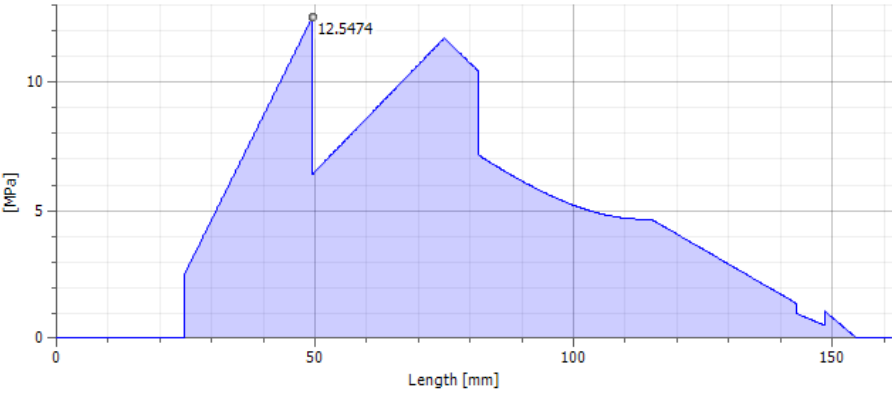
Deflection, YZ Plane



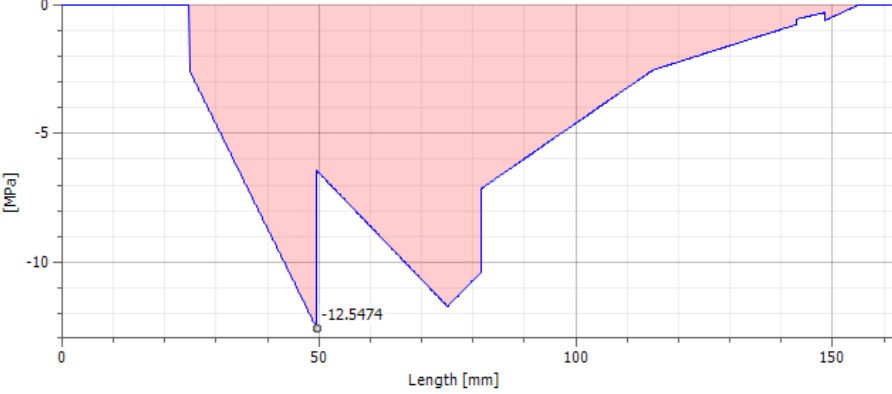
Deflection, XZ Plane



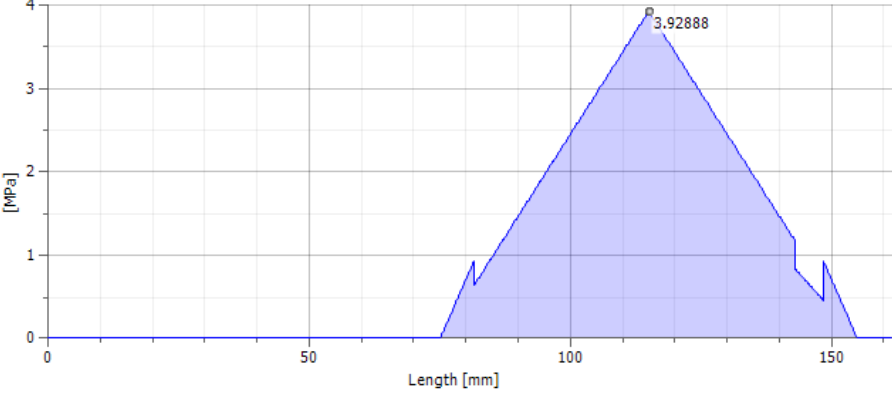
Bending Stress



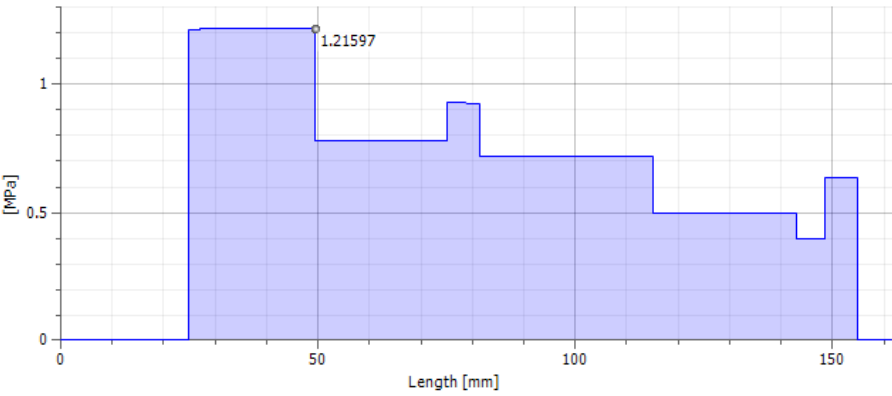
Bending Stress, YZ Plane



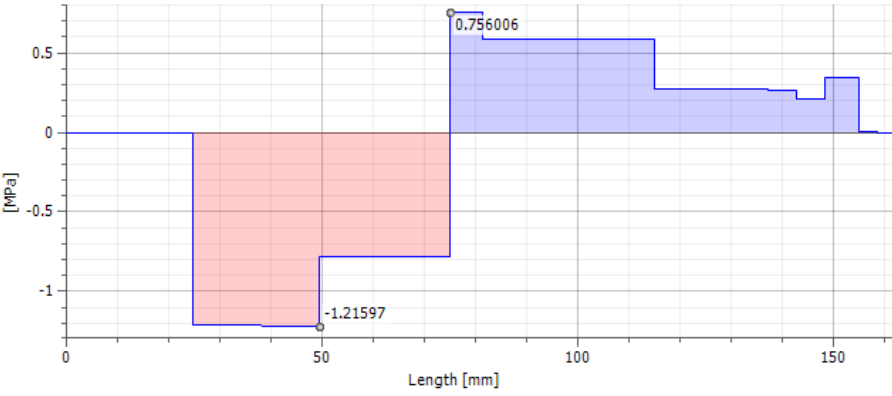
Bending Stress, XZ Plane



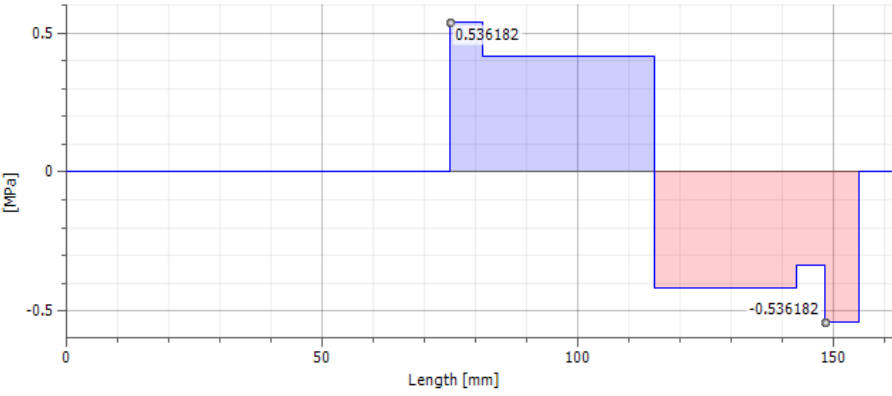
Shear Stress



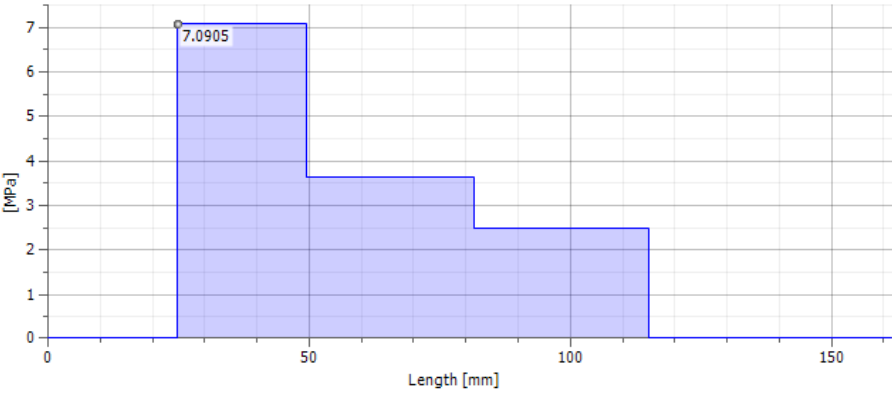
Shear Stress, YZ Plane



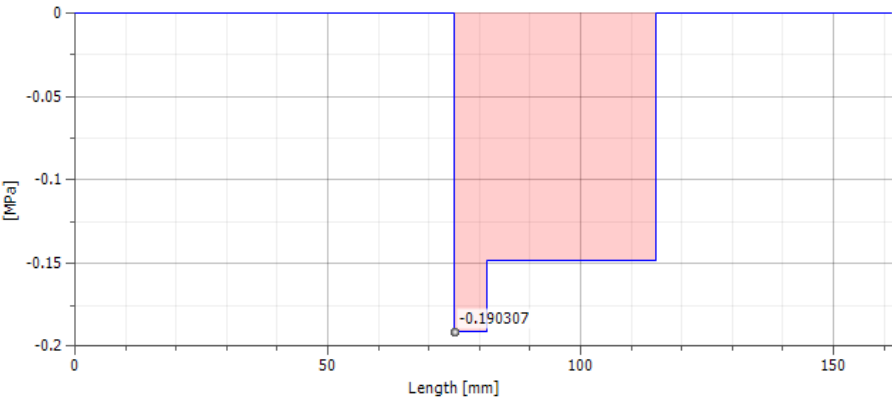
Shear Stress, XZ Plane



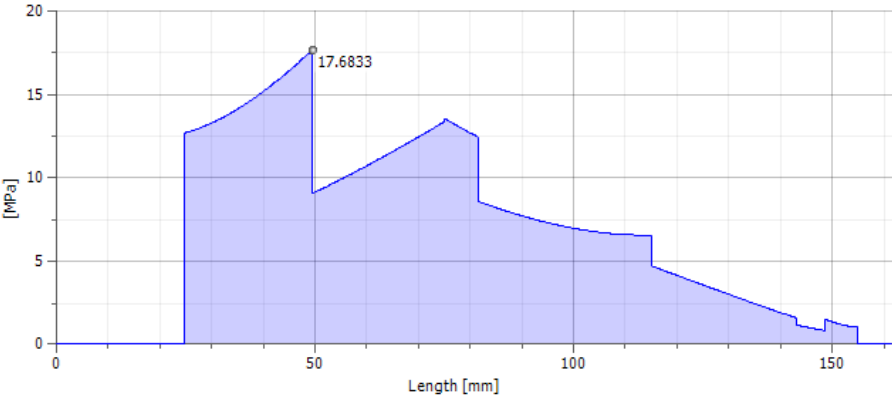
Torsional Stress



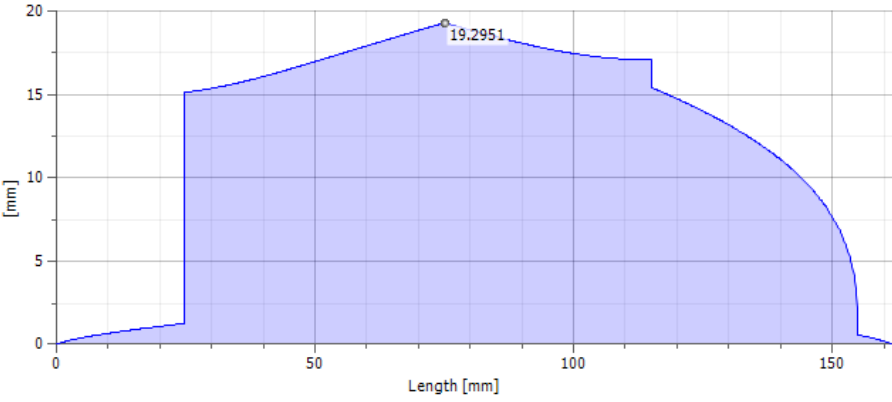
Tension Stress



Reduced Stress



Ideal Diameter



Summary of Messages

11:30:43 AM Calculation: Calculated.

Shaft Component Generator (Version: 2026.2 (Build 302298000, 298))

12/17/2025

Project Info (iProperties)

Calculation

Material

Material		C45
Modulus of Elasticity	E	210000 MPa
Modulus of Rigidity	G	80000 MPa
Density	ρ	7800 kg/m ³

Calculation Properties

Include		
Yes	Density	ρ 7800 kg/m ³
Yes	Shear Displacement Ratio	β 1.188 μ
	Number of Divisions	1000 μ
	Mode of reduced stress	HMH

Loads

Index	Location	Radial Force				Bending Moment				Continuous Load					Axial Force	Torque	Deflection				Deflection Angle
		Y	X	Size	Direction	Y	X	Size	Direction	Y	X	Size	Direction	Length			Y	X	Size	Direction	
1	26.25 mm															77.739 N m	-20.204 microm	0.959 microm	20.227 microm	177.28 deg	0.03 deg
2	26.25 mm					14.070 N m		14.070 N m									-20.204 microm	0.959 microm	20.227 microm	177.28 deg	0.03 deg
3	26.25 mm	1102.120 N		1102.120 N													-20.204 microm	0.959 microm	20.227 microm	177.28 deg	0.03 deg
4	120 mm															-77.739 N m	1.461 microm	-0.700 microm	1.620 microm	334.41 deg	0.00 deg
5	120 mm														134.520 N		1.461 microm	-0.700 microm	1.620 microm	334.41 deg	0.00 deg
6	120 mm	286.560 N		286.560 N													1.461 microm	-0.700 microm	1.620 microm	334.41 deg	0.00 deg
7	120 mm	0.000 N	758.010 N	758.010 N	90.00 deg												1.461 microm	-0.700 microm	1.620 microm	334.41 deg	0.00 deg

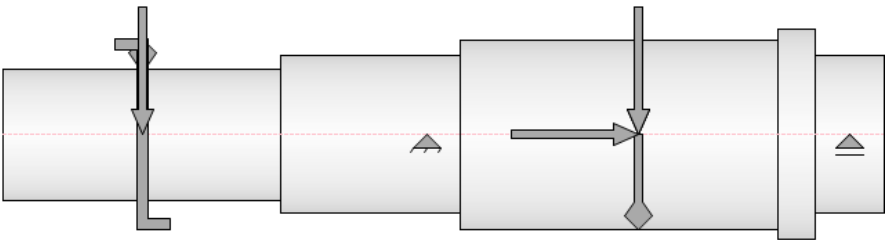
Supports

Index	Type	Location	Reaction Force				Axial Force	Yielding	Type	Deflection				Deflection Angle
			Y	X	Size	Direction				Y	X	Size	Direction	
1	Fixed	80 mm	2169.761 N	379.005 N	2202.614 N	9.91 deg	134.520 N		User	0.000 microm	-0.000 microm	0.000 microm	360.00 deg	0.01 deg
2	Free	160 mm	-771.225 N	379.005 N	859.321 N	153.83 deg			User	-0.000 microm	-0.000 microm	0.000 microm	206.54 deg	0.00 deg

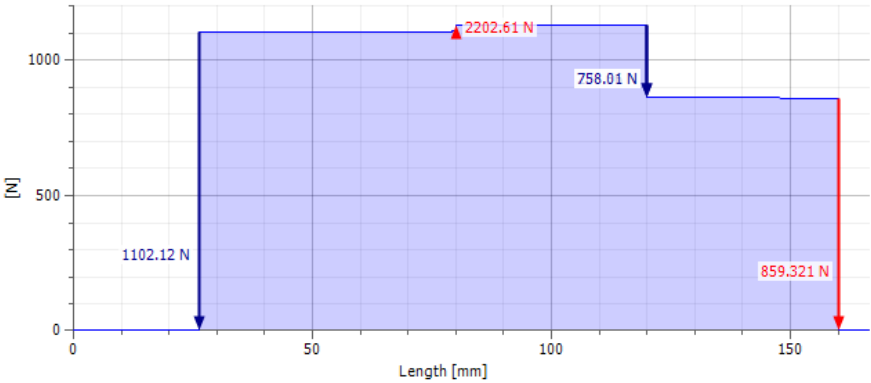
Results

Length	L	166.500 mm
Mass	Mass	1.005 kg
Maximal Bending Stress	σ_B	28.065 MPa
Maximal Shear Stress	τ_S	2.249 MPa
Maximal Torsional Stress	τ	25.339 MPa
Maximal Tension Stress	σ_T	0.190 MPa
Maximal Reduced Stress	σ_{red}	52.240 MPa
Maximal Deflection	f_{max}	34.031 microm
Angle of Twist	ϕ	0.07 deg

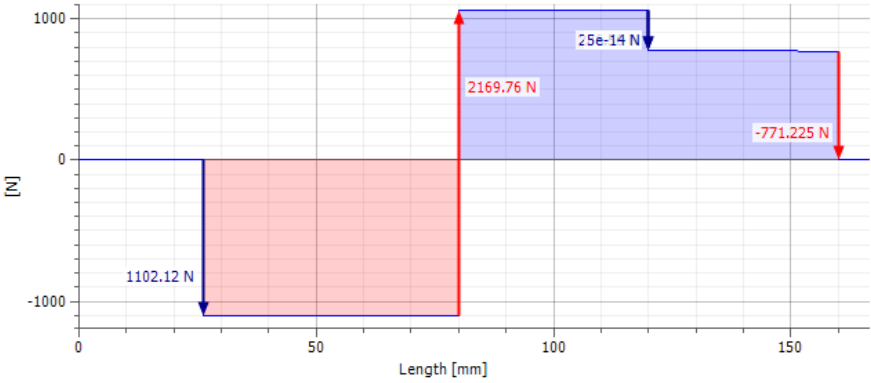
Preview



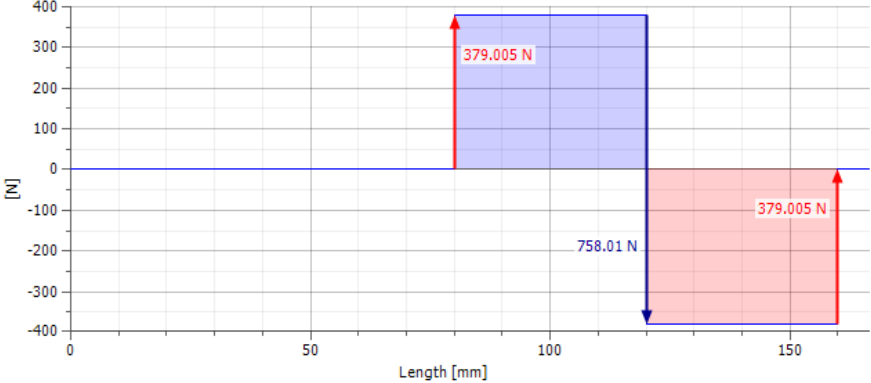
Shear Force



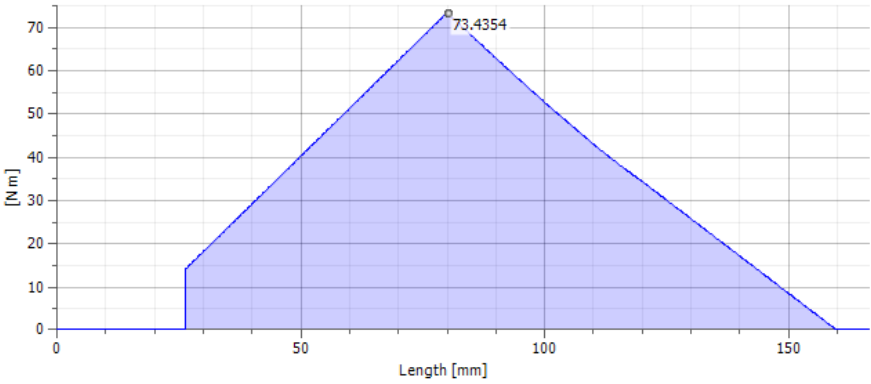
Shear Force, YZ Plane



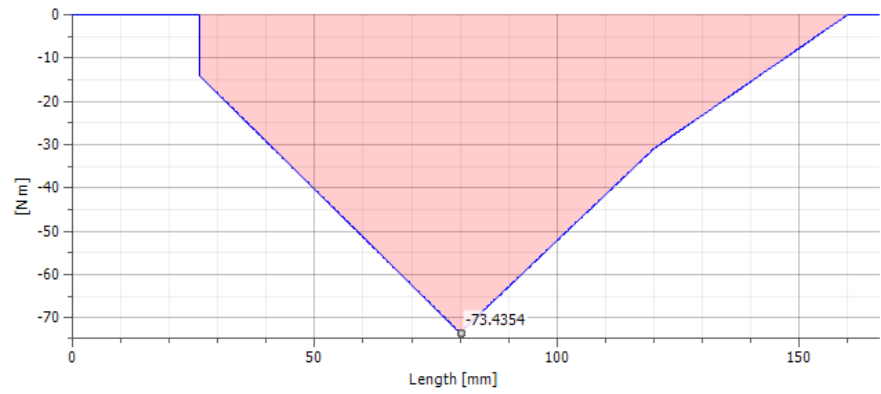
Shear Force, XZ Plane



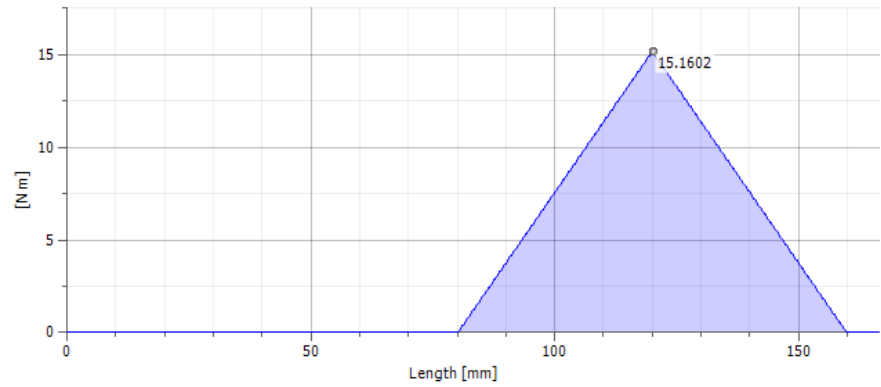
Bending Moment



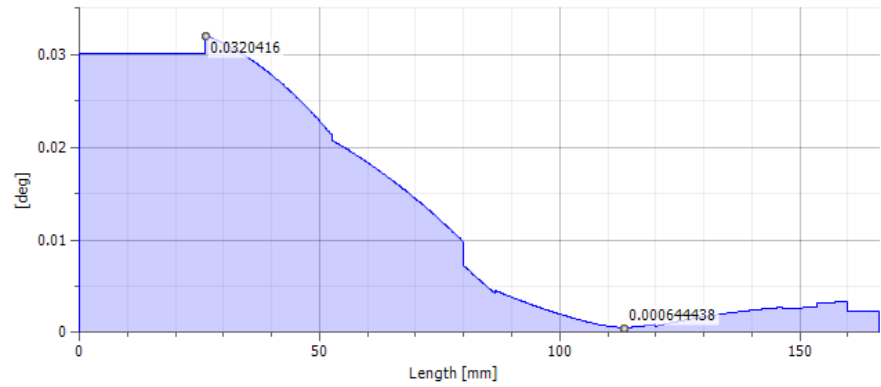
Bending Moment, YZ Plane



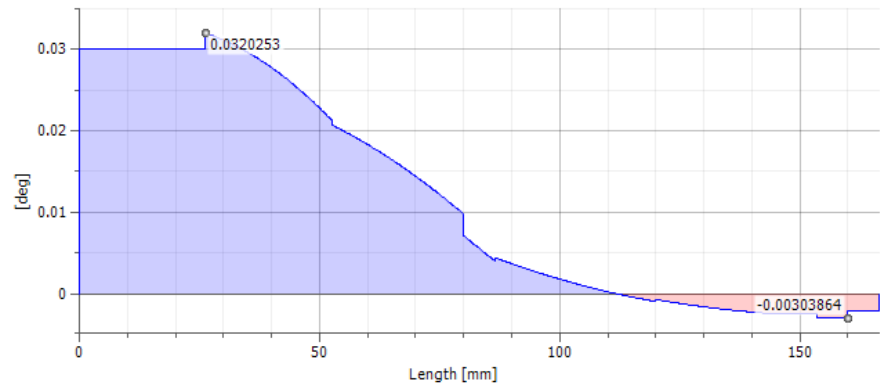
Bending Moment, XZ Plane



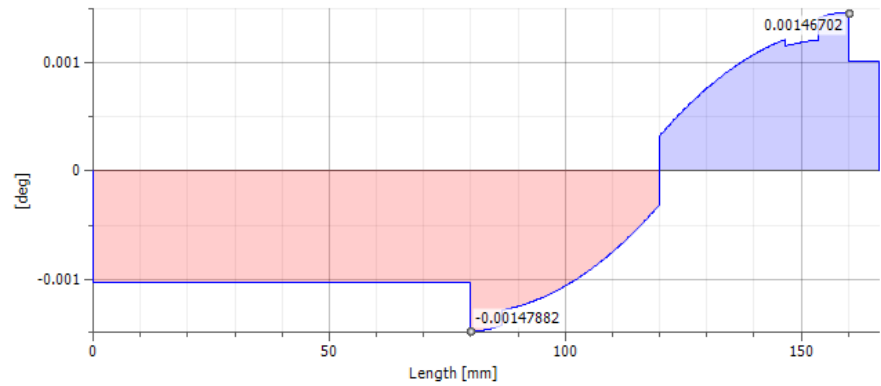
Deflection Angle



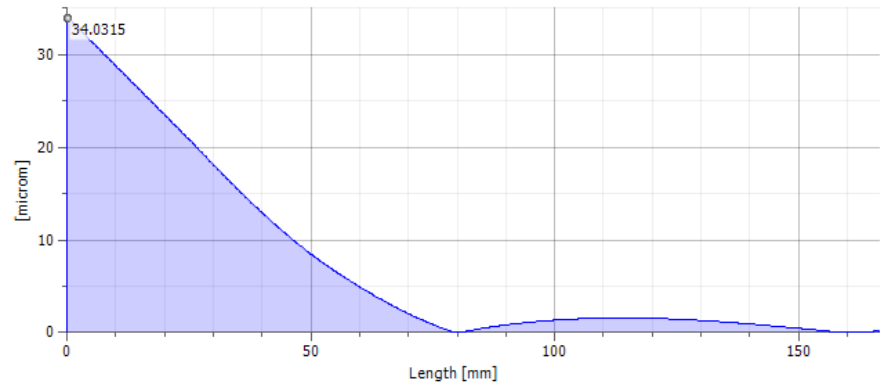
Deflection Angle, YZ Plane



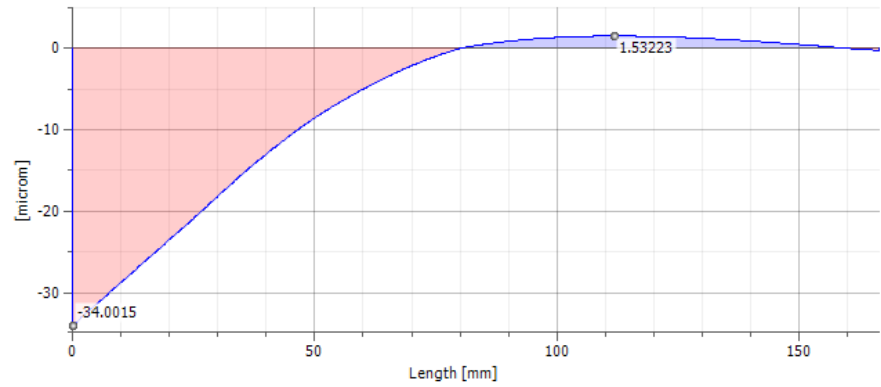
Deflection Angle, XZ Plane



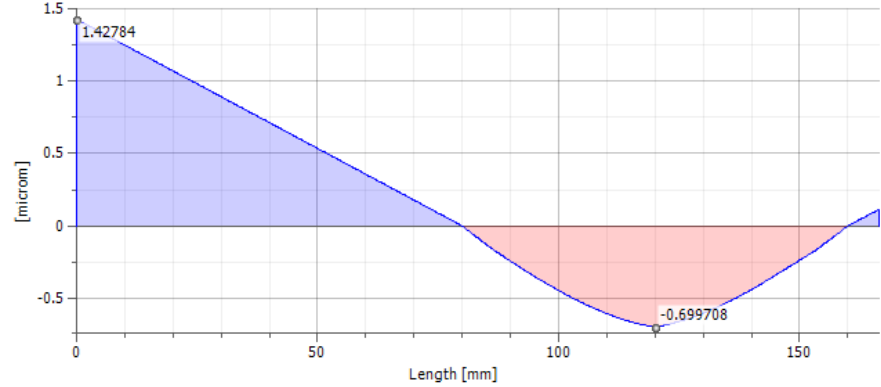
Deflection



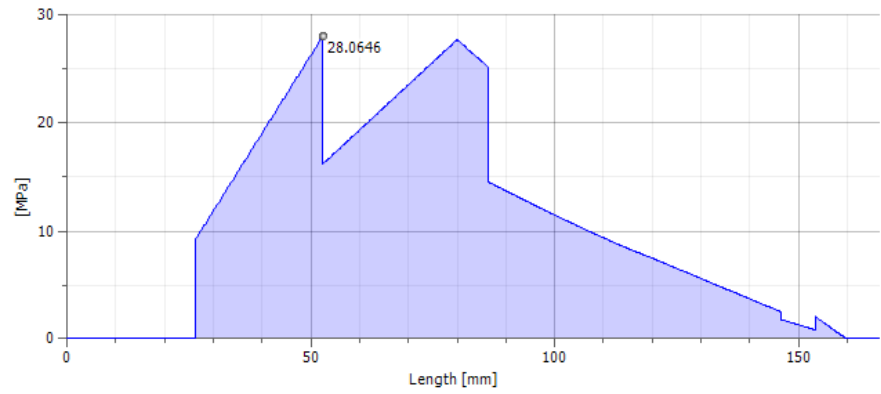
Deflection, YZ Plane



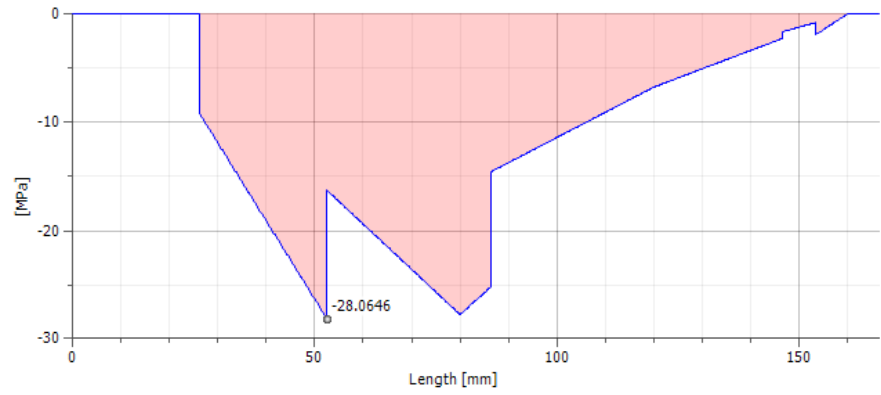
Deflection, XZ Plane



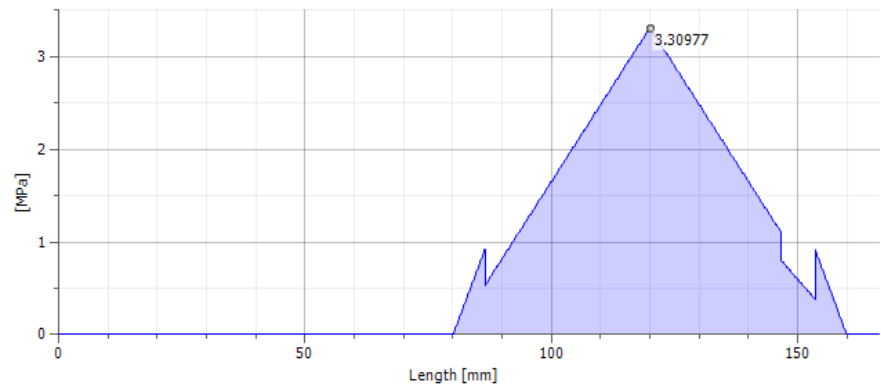
Bending Stress



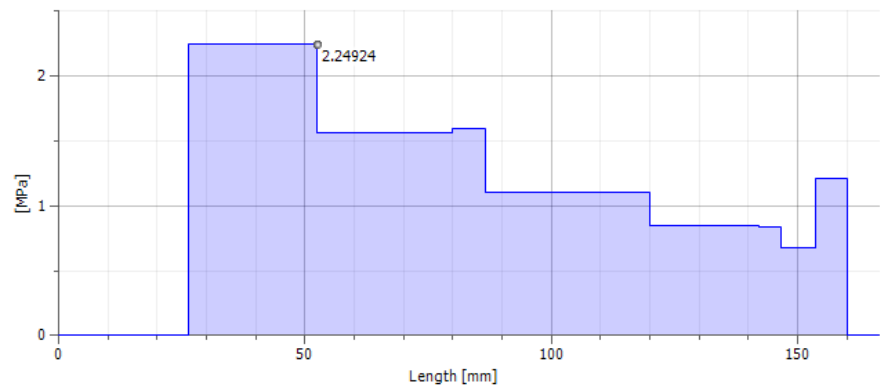
Bending Stress, YZ Plane



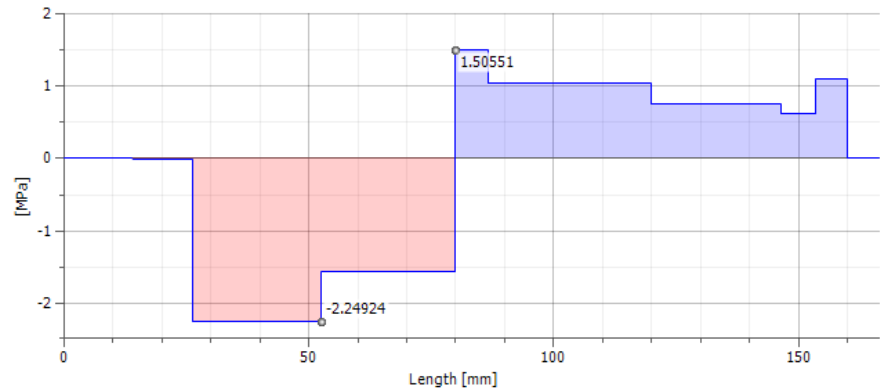
Bending Stress, XZ Plane



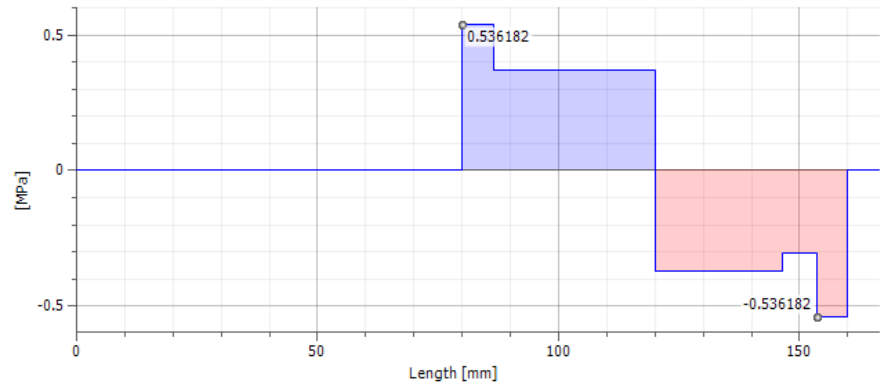
Shear Stress



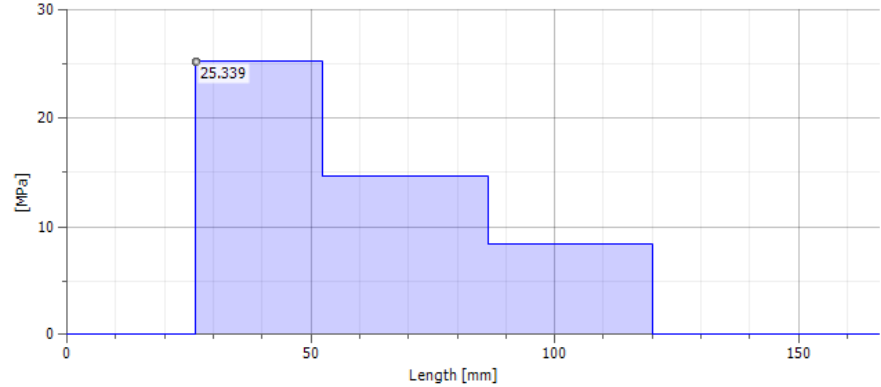
Shear Stress, YZ Plane



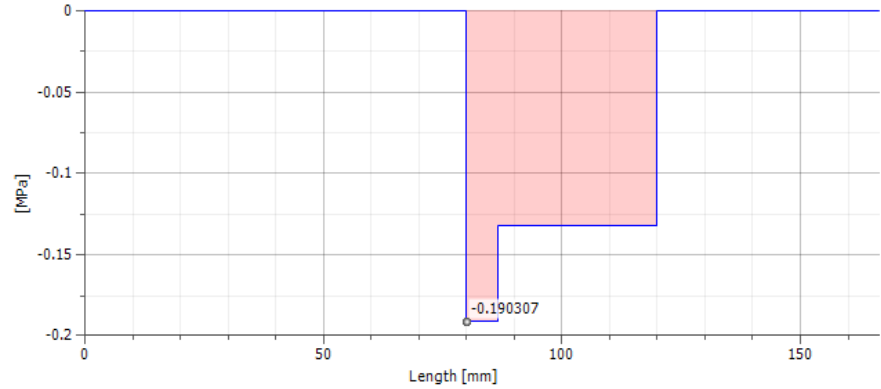
Shear Stress, XZ Plane



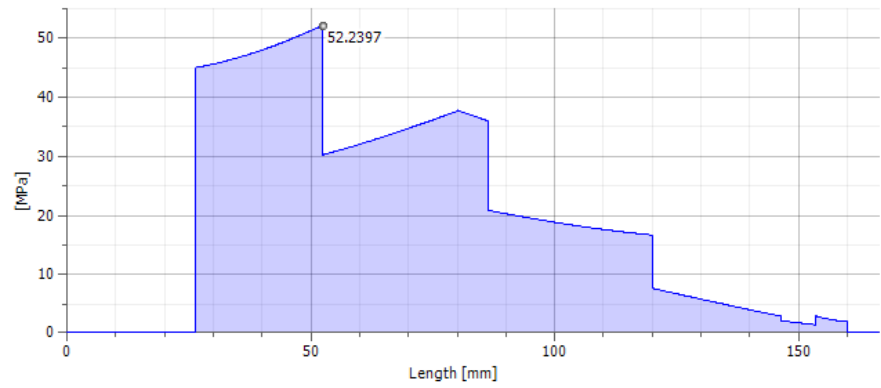
Torsional Stress



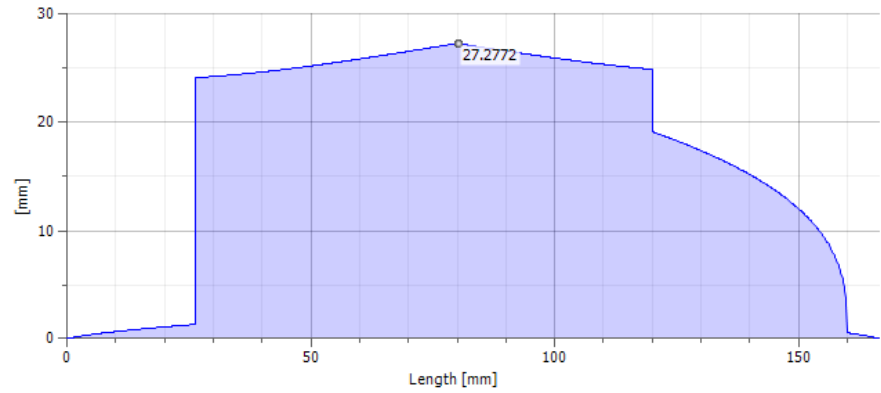
Tension Stress



Reduced Stress



Ideal Diameter



Summary of Messages

11:31:16 AM Calculation: Calculated.

Bearing Component Generator (Version: 2026.2 (Build 302298000, 298))

12/17/2025

Project Info (iProperties)

Guide

Type of Strength Calculation - Check Calculation

Loads

Bearing radial load	F_r	1150 N
Bearing axial load	F_a	135 N
Speed	n	2580 rpm
Required static safety factor	S_0	2.0 ul

Bearing

Designation		DIN 625 SKF (SKF 6006)
Bearing inside diameter	d	30.000 mm
Bearing outside diameter	D	55.000 mm
Bearing width	B	13.000 mm
Nominal contact angle of the bearing	α	5 deg
Basic dynamic load rating	C	13800 N
Basic static load rating	C_0	8300 N
Dynamic radial load Factor	X	1.00 ul / 0.56 ul
Dynamic axial load Factor	Y	0.00 ul / 0.00 ul
Limit value of F_a/F_r	e	0.40 ul
Static radial load Factor	X_0	0.60 ul
Static axial load Factor	Y_0	0.60 ul
Limiting speed lubrication grease	n_{Lim1}	17000 rpm
Limiting speed lubrication oil	n_{Lim2}	28000 rpm

Bearing Life Calculation

Calculation Method		ANSI/AFBMA 9-1990 (ISO 281-1990)
Required rating life	L_{req}	867 hr
Required reliability	R_{req}	90 ul
Life adjustment factor for special bearing properties	a_2	1.00 ul
Life adjustment factor for operating conditions	a_3	1.00 ul
Working temperature	T	100 c
Factor of Additional Forces	f_d	1.00 ul

Lubrication

Friction factor μ	0.0015 ul
Lubrication	Oil

Results

Basic rating life	L_{10}	11160 hr
Adjusted rating life	L_{na}	11160 hr
Calculated static safety factor	s_{0c}	7.21674 ul
Power lost by friction	P_z	6.99146 W
Necessary minimum load	F_{min}	83 N
Static equivalent load	P_0	1150 N
Dynamic equivalent load	P	1150 N
Over-revolving factor	k_n	10.853 ul
Life adjustment factor for reliability	a_1	1.00 ul
Temperature factor	f_t	1.00 ul
Equivalent speed	n_e	2580 rpm
Minimum speed	n_{min}	2580 rpm
Maximum speed	n_{max}	2580 rpm
Strength Check	Positive	

Summary of Messages

11:34:34 AM Calculation: Calculation indicates design compliance!

Bearing Component Generator (Version: 2026.2 (Build 302298000, 298))

12/17/2025

Project Info (iProperties)

Guide

Type of Strength Calculation - Check Calculation

Loads

Bearing radial load	F_r	2203 N
Bearing axial load	F_a	135 N
Speed	n	626 rpm
Required static safety factor	S_0	2.0 ul

Bearing

Designation		DIN 625 SKF (SKF 6006)
Bearing inside diameter	d	30.000 mm
Bearing outside diameter	D	55.000 mm
Bearing width	B	13.000 mm
Nominal contact angle of the bearing	α	5 deg
Basic dynamic load rating	C	13800 N
Basic static load rating	C_0	8300 N
Dynamic radial load Factor	X	1.00 ul / 0.56 ul
Dynamic axial load Factor	Y	0.00 ul / 0.00 ul
Limit value of F_a/F_r	e	0.40 ul
Static radial load Factor	X_0	0.60 ul
Static axial load Factor	Y_0	0.60 ul
Limiting speed lubrication grease	n_{Lim1}	17000 rpm
Limiting speed lubrication oil	n_{Lim2}	28000 rpm

Bearing Life Calculation

Calculation Method		ANSI/AFBMA 9-1990 (ISO 281-1990)
Required rating life	L_{req}	867 hr
Required reliability	R_{req}	90 ul
Life adjustment factor for special bearing properties	a_2	1.00 ul
Life adjustment factor for operating conditions	a_3	1.00 ul
Working temperature	T	100 c
Factor of Additional Forces	f_d	1.00 ul

Lubrication

Friction factor μ	0.0015 ul
Lubrication	Grease

Results

Basic rating life	L_{10}	6548 hr
Adjusted rating life	L_{na}	6548 hr
Calculated static safety factor	s_{0c}	3.76825 ul
Power lost by friction	P_z	3.24881 W
Necessary minimum load	F_{min}	83 N
Static equivalent load	P_0	2203 N
Dynamic equivalent load	P	2203 N
Over-revolving factor	k_n	27.157 ul
Life adjustment factor for reliability	a_1	1.00 ul
Temperature factor	f_t	1.00 ul
Equivalent speed	n_e	626 rpm
Minimum speed	n_{min}	626 rpm
Maximum speed	n_{max}	626 rpm
Strength Check	Positive	

Summary of Messages

11:32:39 AM Calculation: Calculation indicates design compliance!

Parallel Key Component Generator (Version: 2026.2 (Build 302298000, 298))

12/17/2025

Project Info (iProperties)

Guide

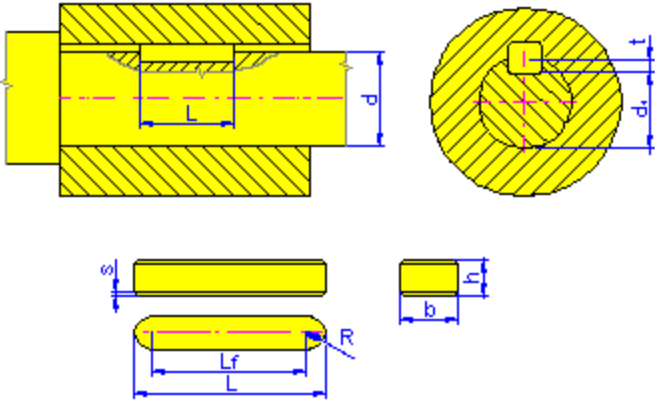
Loads

Dimensions

Shaft Diameter	d	25.00000 mm
Hollow Shaft Inner Diameter	d _h	0.000 mm

Key

Key Width	b	8.000 mm
Key Height	h	7.000 mm
Key Length	l	32.000 mm
Functional Length	l _f	24.000 mm
Key Chamfer	s	0.000 mm
Key Radius	R	0.160 mm



Joint Properties

Number of Keys	N	1.000 ul
Application Factor	K _a	1.000 ul
Fatigue-Life Factor	K _f	1.000 ul
Wear-Life Factor	K _w	1.000 ul
Load Distribution Factor	K _m	1.000 ul
Desired Safety	S _v	2.000 ul

Material

Key

Material	E335
Allowable Pressure	p _A 169.000 MPa

Shaft

Material	C45
Allowable Pressure	p _A 183.000 MPa
Allowable Stress in Shear	τ _A 98.000 MPa

Hub

Material	42CrV6
Allowable Pressure	p _A 280.000 MPa

Results

Min. Functional Key Length	l _{min}	21.028 mm
Min. Shaft Diameter	d _{min}	20.066 mm
Strength Check		Positive

Key

Calculated Pressure	p _c	74.037 MPa
Safety	S	2.283 ul

Shaft

Calculated Pressure	p _c	74.037 MPa
Safety	S	2.472 ul

Hub

Calculated Pressure	p _c	74.037 MPa
Safety	S	3.782 ul

Summary of Messages

11:39:14 AM Calculation: Calculation indicates design compliance!

Parallel Key Component Generator (Version: 2026.2 (Build 302298000, 298))

12/17/2025

Project Info (iProperties)

Guide

Method of Load Calculation - Power calculation for the specified torque and speed
Type of Strength Calculation - Strength check for the specified load, dimensions and joint properties

Loads

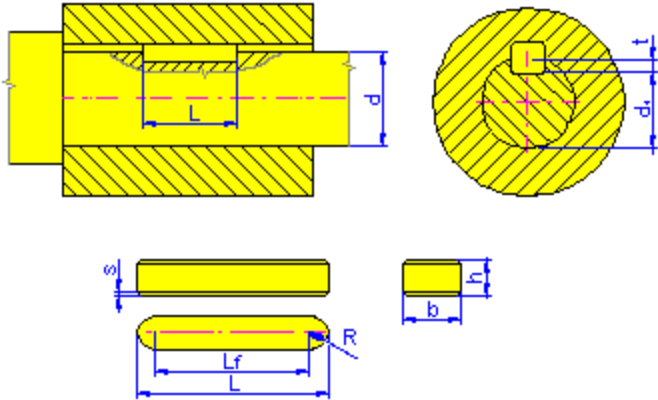
Power	P	5.096 kW
Speed	n	626.000 rpm
Torque	T	77.739 N m

Dimensions

Shaft Diameter	d	36.00000 mm
Hollow Shaft Inner Diameter	d _h	0.000 mm

Key

Key Width	b	10.000 mm
Key Height	h	8.000 mm
Key Length	l	45.000 mm
Functional Length	l _f	35.000 mm
Key Chamfer	s	0.000 mm
Key Radius	R	0.160 mm



Joint Properties

Number of Keys	N	1.000 ul
Application Factor	K _a	1.000 ul
Fatigue-Life Factor	K _f	1.000 ul
Wear-Life Factor	K _w	1.000 ul
Load Distribution Factor	K _m	1.000 ul
Desired Safety	S _v	2.000 ul

Material

Key

Material	E335	
Allowable Pressure	p _A	169.000 MPa

Shaft

Material	C45	
Allowable Pressure	p _A	183.000 MPa
Allowable Stress in Shear	T _A	98.000 MPa

Hub

Material	42CrV6	
Allowable Pressure	p _A	280.000 MPa

Results

Min. Functional Key Length	$l_{\min}$	12.778 mm
Min. Shaft Diameter	$d_{\min}$	20.066 mm
Strength Check	Positive	

Key

Calculated Pressure	p_c	30.849 MPa
Safety	S	5.478 ul

Shaft

Calculated Pressure	p_c	30.849 MPa
Safety	S	5.932 ul

Hub

Calculated Pressure	p_c	30.849 MPa
Safety	S	9.077 ul

Summary of Messages

11:39:51 AM Calculation: Calculation indicates design compliance!